

Marvin Raditya Nugraha

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A third-year Informatics Engineering student at Politeknik Negeri Jakarta with a robust foundation in Data Science, Machine Learning, and Full-Stack Development. Proven ability to architect end-to-end solutions—from engineering predictive maintenance models to building scalable web platforms and IoT integrations. Experienced in leading Agile teams, I am highly passionate about bridging complex data structures with intuitive system designs to solve real-world problems and drive actionable insights.

EDUCATION

Politeknik Negeri Jakarta

2023 - 2027 (Expected)

D-IV (Applied Bachelor's) Informatics Engineering | GPA: 3.92/4.00

Interest Areas: Data Science, Machine Learning Engineering, Full-Stack Development

- Relevant Coursework: Artificial Intelligence, Advanced Web Programming, Algorithms & Data Structures, Database System, Data Mining

WORK EXPERIENCE

Data Verifier (Digitalization Team) | Kementerian Pendidikan Dasar dan Menengah

Sept 2025 – June 2026

- Maintained and enhanced an existing enterprise web application, consistently deploying new features and ensuring system stability to support daily departmental workflows.
- Resolved critical system bugs and data synchronization issues, successfully fixing a data duplication error where processed items redundantly appeared in pending queues.
- Conducted regular system troubleshooting by analyzing data discrepancies and refining logic to align with updated administrative requirements.

IT Support Intern | Pusat Pelatihan Kerja Pengembangan Industri (PPKPI) Jakarta

Feb 2022 – Apr 2022

- Provided first-level technical support to trainees and staff on issues related to computer hardware and internet connectivity.
- Performed troubleshooting and resolved problems with network connectivity and system configurations to ensure smooth daily operations.
- Designed and created visual content, such as posters and infographics, for the official Instagram account @ppkpijakartaofficial.

PROJECTS

PRISM (Predictive Reporting & Intelligent Scheduling Manager) [API Repo](#) • [Frontend Repo](#).

Tech Stack: Python, Scikit-Learn, Laravel, PostgreSQL, React

- Led an industrial Project-Based Learning (PBL) initiative utilizing real-world operational data from PT Nusa Tekno Global to develop an intelligent system predicting optimal maintenance schedules and minimizing unexpected asset downtime.
- Engineered a predictive model using machine learning algorithms to analyze the company's historical asset performance, successfully identifying actual degradation patterns.
- Designed an automated reporting dashboard that translates complex predictive insights into actionable maintenance alerts for the management team.

TagiT – Location-Based Reminder Application [\[Github\]](#)

Tech Stack: Flutter, Firebase, Riverpod, Geofencing API

- Developed a cross-platform mobile application designed to trigger precise, location-aware user reminders by utilizing advanced geofencing technology.
- Engineered a robust background notification system that seamlessly integrates custom alarms, device vibration, and Text-to-Speech (TTS) functionalities to alert users upon entering specific geographical zones.
- Implemented efficient state management using Riverpod and designed a highly interactive user interface, featuring map-based editing and drag-and-drop location tagging.

Classroom Monitoring System (Behavior & Cheating Detection) [\[Github\]](#)

Tech Stack: YOLO, Python, TensorFlow, Keras, OpenCV

- Developed a computer vision-based monitoring system designed to automatically track and classify student behaviors, specifically targeting the detection of academic dishonesty during examinations.
- Designed and trained a custom Convolutional Neural Network (CNN) architecture to extract and analyze complex visual features from classroom environmental data.
- Applied advanced image preprocessing and data augmentation techniques to handle varying real-world classroom conditions (e.g., lighting, camera angles) and improve the model's robustness on imbalanced datasets.

CenPi (Central Canteen Payment Integration) – Canteen Digitalization App [\[Github\]](#).

Tech Stack: PHP, Oracle SQL, HTML, CSS, JavaScript, Midtrans API, WebSockets, Cloudflare, Tailscale

- Acted as Scrum Master, guiding the team using Agile methodologies, ensuring the smooth execution of 4 sprint cycles, and removing impediments to maintain team productivity.
- Implemented design mockups into functional and interactive frontend user interfaces.
- Developed the core backend business logic for key features, including CRUD operations, an analytics dashboard, and database connectivity.

LEADERSHIP & ACHIEVEMENTS

EXPECTIK (Exploration and Perception Week)

Sept 2025 – Nov 2025

Head of Academic Division

- Directed and managed a cross-functional team of 7 staff members, overseeing the end-to-end planning and execution of the orientation program for over 300+ new Informatics Engineering students.
- Spearheaded the curriculum development strategy, ensuring all academic, creative, and self-development assignments aligned with the department's core educational objectives.
- Facilitated conflict resolution and team synergy, acting as the primary decision-maker to maintain smooth operational workflows and high team productivity throughout the event.

Career Expo Committee Politeknik Negeri Jakarta

Jul 2025 – Aug 2025

Security & Logistics Division Staff

- Managed all logistical and operational aspects of the event, including equipment procurement, vendor contracts, consumption distribution, and first aid readiness.
- Ensured event security and order by identifying potential risks and proactively supervising all activities during the event.

EXPECTIK (Exploration and Perception Week) Committee

Sept 2024 – Nov 2024

Academic Division Staff

- Designed and developed a comprehensive assignment curriculum for over 300 new students, covering academic, self-development, and creative aspects.
- Actively contributed to creating a positive and substantive orientation experience to facilitate the smooth adaptation of new students into the Informatics Engineering department.

Guest Speaker | **New Student Welcome TIK 2024**

August 2024

- Specially invited as the department's top-performing student to be a speaker at the new student orientation event.
- Shared insights and delivered material on academic success tips, time management, and adaptation strategies for university life.

Dean's List (**Outstanding Student Award**)

- Achieved the highest semester Grade Point Average (GPA) for Fall 2023 in the Informatics Engineering department

CERTIFICATIONS

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- [HCIA-AI V4.0 Course](#) | *Huawei ICT Academy* | 2026
 - [Alibaba Cloud Certified Associate](#) – Cloud Computing | *Alibaba Cloud* | 2024
 - [Data Science & SQL Fundamentals](#) | *Dicoding Indonesia* | 2024
 - [Level II Competency Certification in Computer Network Engineering](#) | *Indonesian National Board for Professional Certification (BNSP)* | 2023
 - [Python](#), [Machine Learning](#), & [Data Visualization](#) Fundamentals | *Dicoding Indonesia* | 2023

SKILLS

Programming Languages: Python, JavaScript/TypeScript, PHP, Dart, SQL

Machine Learning & Data Science: TensorFlow, Keras, Scikit-Learn, OpenCV, Computer Vision

Web & Mobile Frameworks: React / Next.js, Laravel, Flutter, Tailwind CSS.

Databases & Backend: PostgreSQL, Prisma ORM, Firebase, RESTful API Architecture.

Tools: Canva, Figma, Office Suite (Word, Excel, PowerPoint), WordPress, VSCode, Github, Notion